Risk vs Return of 8 Dhaka Stock Exchange Stocks

24 Dec 2020 to 8 Apr 2025 | Excel workbook with live formulas | Prepared by Muhammad Sadat Samin

Question

Which of 8 DSE stocks, and which of 4 sectors, paid best for the risk taken? And can a lower-risk portfolio be built from them?

Data and cleaning

Source: Mendeley Data, *Dhaka Stock Exchange Historical Data (1999-2025)*, Sunny, Nafis and Khan (2025), CC BY 4.0, DOI 10.17632/5mww8rb9td.1. Stocks: GP and ROBI (telecom), SQURPHARMA and RENATA (pharma), BRACBANK and ISLAMIBANK (banks), BATBC and UNILEVERCL (consumer). The raw file needed work. 184 trading days between 18 Jul 2022 and 16 Apr 2023 appeared twice, and many dates before July 2022 had day and month swapped, which would have produced 135 false crashes of more than 10%. I removed the duplicates and fixed the dates. A day with no trade uses the previous close. ROBI only listed on 24 Dec 2020, so all 8 stocks are measured from that date: 1,031 trading days and 1,030 daily returns.

Method

Daily simple returns, 250 trading days a year, and an 8% annual risk-free rate. The 8% is a placeholder I chose, not a Bangladesh Bank figure, and it can be changed in one cell. Annual return is the compound rate (CAGR). Volatility is the standard deviation of daily returns times the square root of 250. Sharpe ratio is (CAGR minus 8%) divided by volatility. Max drawdown is the biggest fall from an earlier peak. The workbook also has an 8 x 8 correlation matrix and sector averages. The minimum-variance weights come from an optimizer (long-only, summing to 100%) and are typed in as inputs; SUMPRODUCT and MMULT formulas then compute that portfolio's return and risk.

Stock / portfolio	Sector	Price, start to end (BDT)	Total return	CAGR	Volatility	Sharpe	Max drawdown	Min-var weight
ROBI	Telecom	15.00 to 26.00	+73.33%	+14.28%	43.91%	+0.14	-70.76%	0.00%
ISLAMIBANK	Banks	26.70 to 43.30	+62.17%	+12.45%	25.57%	+0.17	-38.92%	10.17%
BRACBANK	Banks	44.30 to 49.00	+10.61%	+2.48%	24.99%	-0.22	-44.76%	0.00%
SQURPHARMA	Pharma	207.60 to 216.20	+4.14%	+0.99%	13.24%	-0.53	-20.31%	48.45%
GP	Telecom	341.10 to 320.50	-6.04%	-1.50%	17.69%	-0.54	-42.70%	12.62%
UNILEVERCL	Consumer	2,927.60 to 2,411.40	-17.63%	-4.60%	30.55%	-0.41	-46.83%	6.00%
RENATA	Pharma	1,080.50 to 496.50	-54.05%	-17.20%	17.54%	-1.44	-66.37%	22.76%
BATBC	Consumer	1,078.40 to 321.80	-70.16%	-25.44%	39.35%	-0.85	-81.42%	0.00%
Equal-weight	8 stocks, 12.5% each	100.00 to 100.91	+0.91%	+0.22%	14.69%	-0.53	-38.59%	12.50% each
Min-variance	5 stocks	100.00 to 91.86	-8.14%	-2.04%	10.69%	-0.94	-31.06%	see note

Prices are not adjusted for bonus shares or cash dividends (see Limitations). Min-variance weights: SQURPHARMA 48.45%, RENATA 22.76%, GP 12.62%, ISLAMIBANK 10.17%, UNILEVERCL 6.00%; ROBI, BRACBANK and BATBC get 0%.

Findings

Only two stocks beat the 8% risk-free rate. ISLAMIBANK returned 12.45% a year with a Sharpe ratio of +0.17, ROBI 14.28% with +0.14. ROBI's gain starts from its 15 BDT listing-day price. It peaked at 70.1 BDT on 14 Jan 2021 and its worst drawdown was -70.76%, so I would not call it a clear winner. SQURPHARMA was the calmest stock (13.24% volatility, worst drawdown -20.31%) and gained 4.14% in total.

Sectors. Banks came out best on average (total return 36.39%, CAGR 7.46%, Sharpe -0.02), then Telecom (33.65%). Pharma had the lowest volatility (15.39%) but averaged -24.95% because of RENATA. Consumer averaged -43.90%, and that figure is distorted by unadjusted prices.

Diversification helped risk, not return. The average correlation between the 8 stocks is 0.203, from 0.083 (ISLAMIBANK and UNILEVERCL) to 0.391 (GP and BRACBANK). The equal-weight portfolio has 14.69% volatility, and 100 BDT became 100.91 (100.30 if never rebalanced). The minimum-variance portfolio cut volatility to 10.69%, which is about 2.5 percentage points below the safest single stock. But it only minimizes risk and ignores return: 100 BDT became 91.86, and its Sharpe ratio (-0.94) is worse than equal weight (-0.53). Both portfolios have negative Sharpe ratios.

Three one-day drops of 10% or more. BATBC fell 64.17% on 4 Mar 2021, in line with its 200% stock dividend (the share count tripled). UNILEVERCL fell 37.46% on 4 Apr 2023 after a 60% stock dividend: 3,200.2 divided by 1.6 is 2,000.1, against an actual close of 2,001.5. ROBI fell 10.00% on 19 Mar 2024 (30 to 27 BDT). A 1 BDT cash dividend cannot explain a 3 BDT fall, so this is only partly explained, probably the daily price limit. I flagged these drops and did not adjust the prices.

Limitations

(1) Prices are not adjusted for bonus shares or cash dividends. BATBC, RENATA and UNILEVERCL returns are understated, and much of BATBC's -70% comes from its 2021 stock dividend, not a real loss of that size. (2) BSEC floor prices from Jul 2022 to Jan 2024 froze many stocks for months (GP at 286.6, BATBC at 518.7, SQURPHARMA at 209.8 BDT), which understates volatility. (3) ROBI has a shorter history and a listing-day spike that inflates its volatility and drawdown. (4) Filling no-trade days with the previous close adds zero-return days and slightly lowers measured volatility. (5) The 8% risk-free rate is a placeholder, and changing it changes every Sharpe ratio. (6) Eight stocks over about four years is a small sample. This is a learning project, not investment advice.